CHINA · ROBOTICS · PHYSICAL AI

CHINESE ROBOTICS AND EMBODIED AI

Robots, physical intelligence, factories and productive transformation

Sergio Gevatschnaider



BLOCK 01 / 07

Industrial landscape

Scale and operational capability

What does robot adoption actually tell us?

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Automation at scale

Distinguishing installations, operational stock and robot density

354,000

industrial robots installed in China in 2025

59%

of global installations in 2025

55%

of new Chinese installations supplied by domestic manufacturers

Source: IFR, World Robotics 2026. These are industrial robots, not humanoid counts.

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Three markets with distinct characteristics

Industrial robots, service robots and humanoids are different categories

01

Industrial

Manipulators for welding, palletising, assembly and repetitive operations.

02

Service

Logistics, cleaning, inspection and assistance outside the industrial robot category.

03

Humanoid

An anthropomorphic configuration offering mobility and manipulation possibilities.

A humanoid shape alone does not establish general autonomy.

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The transition to physical AI

Capabilities that make a robot an adaptive system



Learning may take place outside the real-time control loop.

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Evidence behind the growth

What the indicators establish and what they leave open

01

Installations

An annual flow that responds to the investment cycle.

02

Operational stock

The accumulated operating fleet, distinct from annual additions.

03

Density

Operational robots per 10,000 manufacturing employees.

04

Productivity

It need not increase in proportion to robot stock.

Installations, effective automation and aggregate productivity measure different things.

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The strategic question

Why China combines scale, suppliers and industrial policies

A

Ecosystem strengths

Proximity among integrators, motors, sensors, manufacturers, software and domestic demand.

B

Persistent constraints

Safety, robustness, task data, integration costs and investment returns.

Performance depends on the technical and organisational system surrounding the robot.

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BLOCK 02 / 07

Robot architecture

Perception, planning and control

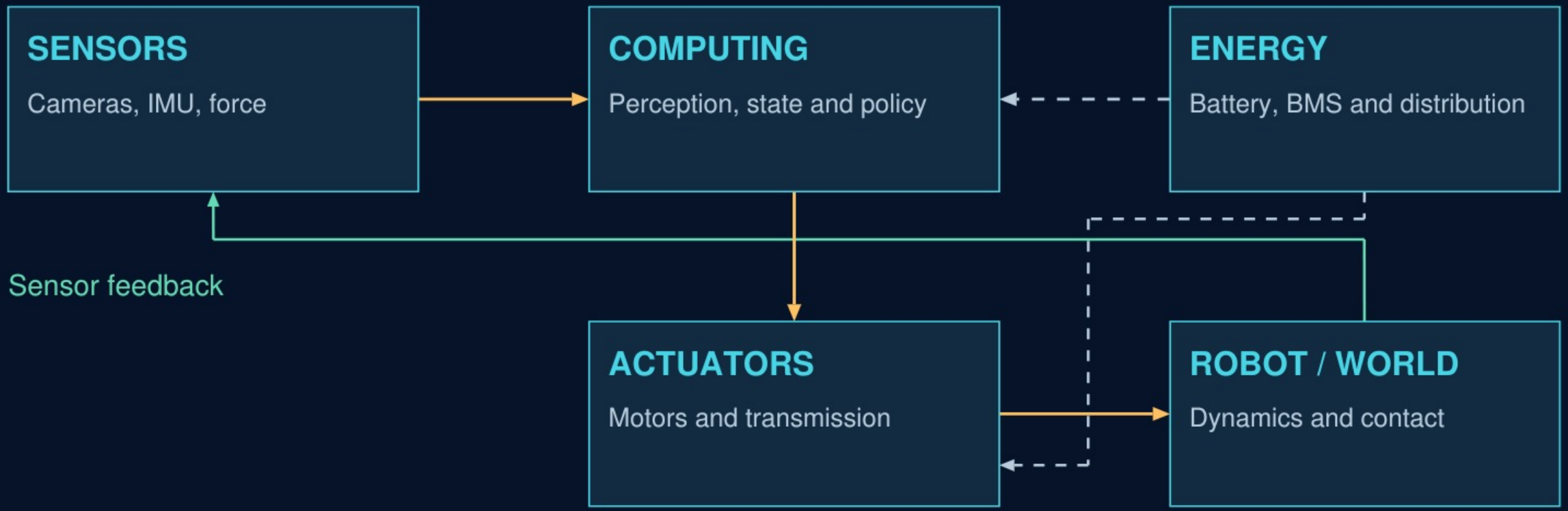
How does an instruction become a verifiable physical action?

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Functional anatomy of a robot

Sensors, actuators, computing, energy and control

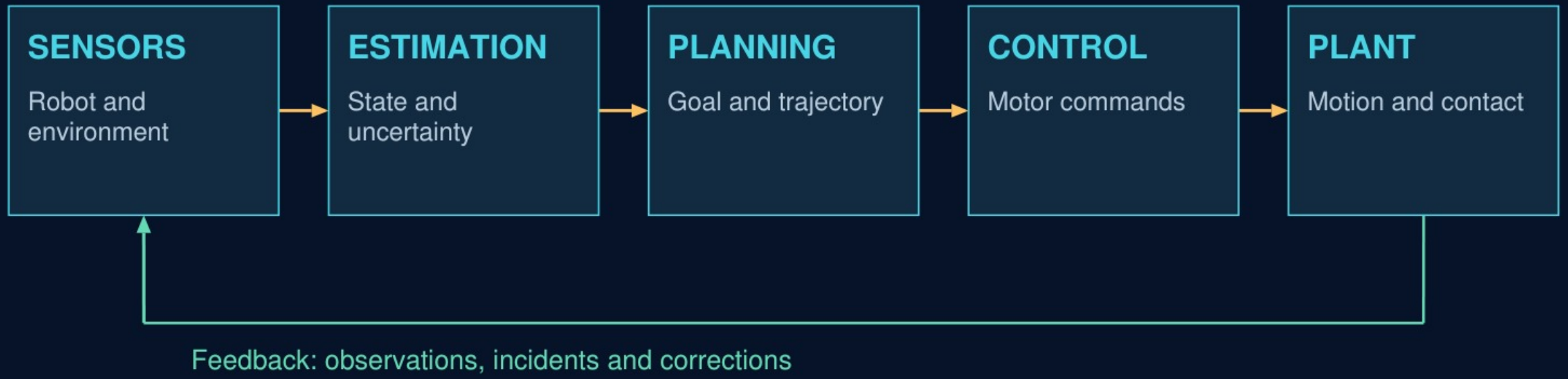


Computing directs actuation. Energy and mechanics constrain the task.

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Closed-loop control

The physical world requires continuous observation and correction

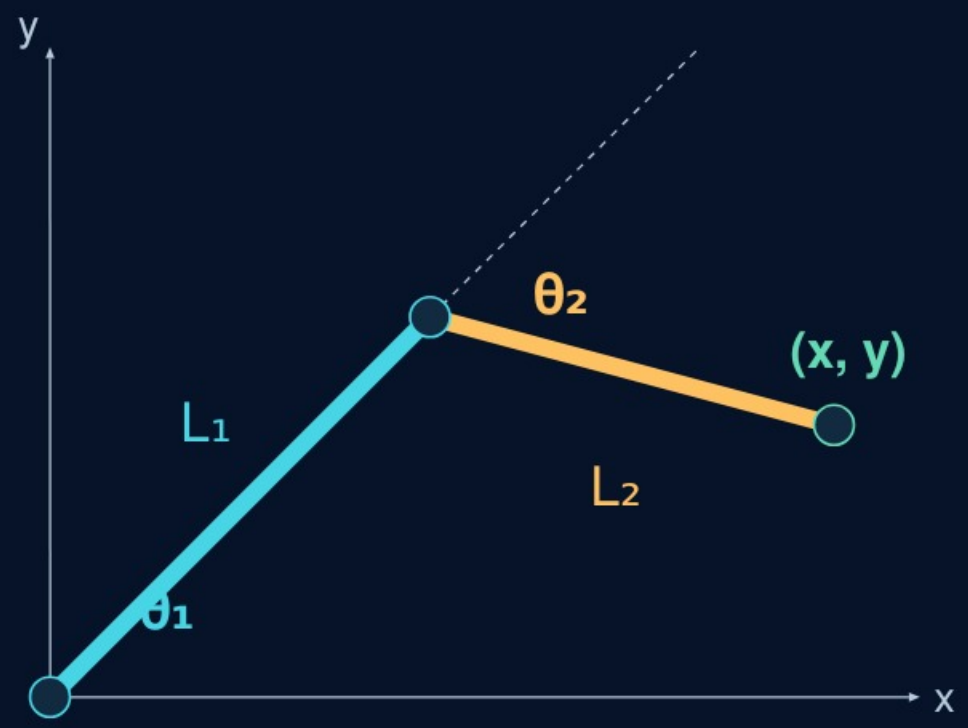


The return arrow closes the physical observation–action loop.

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Forward and inverse kinematics

Joint angles and link lengths determine positions in space



FORWARD

$$x = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$
$$y = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

INVERSE

Given (x, y), find feasible angles.
There may be several solutions or none.

Connected 2R arm. θ_2 is relative to the first link, rather than the x-axis.

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Multimodal perception and localisation

Robotics under imperfect observation

01

Vision

Detection, segmentation and depth.

02

Proprioception

IMU, encoders and joint state.

03

Force and touch

Contact, grasping and slip.

04

Fusion

Latent state, uncertainty and calibration.

Sensor errors propagate into plans and may require stopping a motion.

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Motion planning

A trajectory that can execute the task

1	**State**	Estimate robot and object poses.
2	**Goal**	Define the task, tolerances and constraints.
3	**Plan**	Find a collision-free trajectory.
4	**Control**	Execute with feedback and physical limits.
5	**Validation**	Check the result or replan.

Planning, control and safety have complementary roles.

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BLOCK 03 / 07

Embodied intelligence

VLMs, VLAs and action policies

What changes when foundation models control robots?

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Language and physical action

LLMs, VLMs and VLAs: similarities and differences



Language and observations enter the policy. The environment returns observations.

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Imitation, reinforcement and teleoperation

Complementary sources of learning

01

Demonstrations

Teleoperation and expert data for imitation and behaviour cloning.

02

Simulation

Domain randomisation and policy training with feedback.

03

Deployment

Validation, failure data, adaptation and monitoring.

Learning a skill does not guarantee transfer to another scene.

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Sim-to-real: the reality gap

Why a policy trained in simulation can fail

01

Dynamics

Imperfect models of mass, friction and deformation.

02

Perception

Noise, occlusion, reflections and latency.

03

Environment

New objects, people and unexpected changes.

04

Safety

Limits on force, speed and separation.

Simulation reduces experimental costs. Physical testing remains necessary.

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Generalising a skill

A successful demonstration does not establish operational reliability

Variation

Changes in objects, lighting and position.

Disturbance

Impacts, unexpected motion and failures.

Horizon

Long tasks accumulate opportunities for error.

Evaluation needs repeated trials and tests outside the training distribution.

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A model within a safe system

A layered architecture for physical tasks



Functional schematic: the learned policy requires validated safety functions.

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BLOCK 04 / 07

China's robotics ecosystem

Motors, sensors and humanoid platforms

Which specialisations form the domestic supply chain?

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China's competitive landscape

Distinguishing platforms, components and integrators

01

Unitree

Locomotion, quadruped platforms and humanoids.

02

UBTECH

Humanoids for pilots and industrial tasks.

03

AGIBOT / Zhiyuan

Development and demonstrations of embodied robots.

04

Regional suppliers

Integrators, actuator manufacturers and sensor suppliers.

Marketing claims need comparison with repeatable operations.

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Unitree: locomotion and dexterity

A motion demonstration addresses only part of the problem

A

Demonstrated abilities

Dynamic mobility, balance and coordination across degrees of freedom.

B

Further evaluation

Repeatable industrial manipulation, tolerances, safety and cost per useful operation.

Acrobatics alone do not establish productive performance in a factory.

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UBTECH: industrial pilots

Walker S, Walker S1/S2 and experiments in factories

1	**Environment**	Identify the task and traffic rules.
2	**Perception**	Locate parts, people and stations.
3	**Manipulation**	Handle, sort or inspect.
4	**Coordination**	Interact with cells and other robots.
5	**Validation**	Measure uptime, quality and human intervention.

Pilots show feasibility. Profitability at scale requires further evidence.

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AGIBOT: demonstrations and production

An emerging generation of physical tasks and data

01

Demonstrations

Visual evidence of specific tasks.

02

Production line

Repeated operation, integration and quality control.

03

Scale

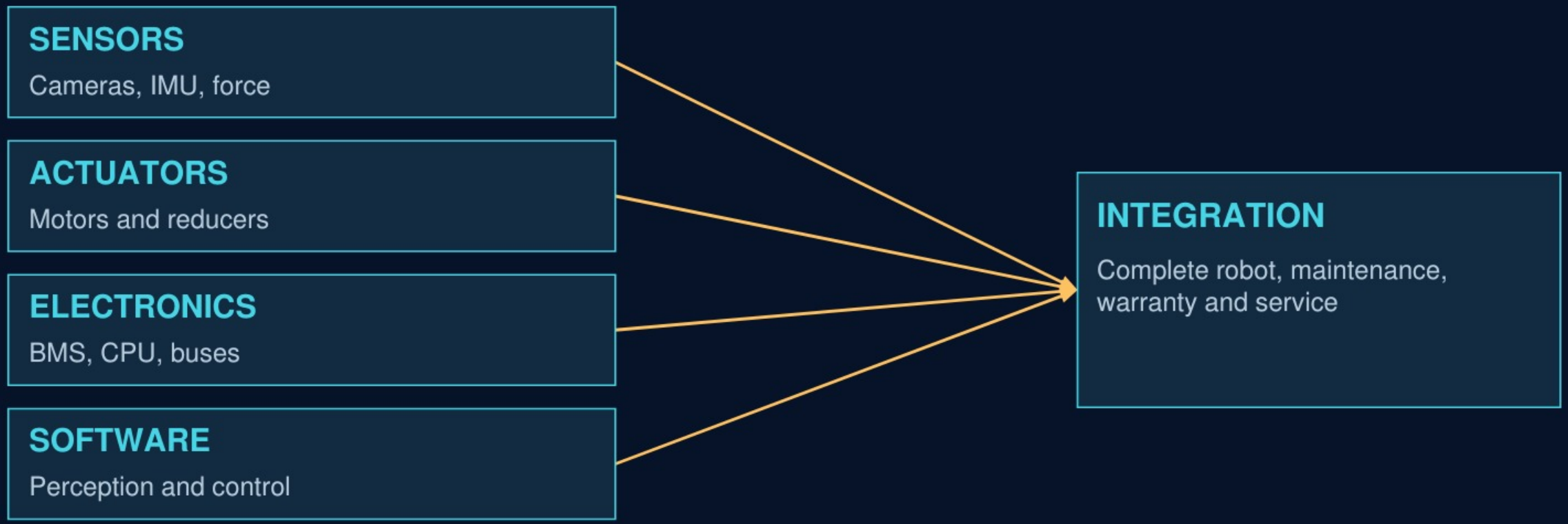
Data on units, support and reliability.

Distinguish announcements, demonstrations and audited field data.

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The component supply chain

Industrial structure of the humanoid ecosystem



Complementary components converge on system integration.

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BLOCK 05 / 07

The smart factory

A system of machines, people and data

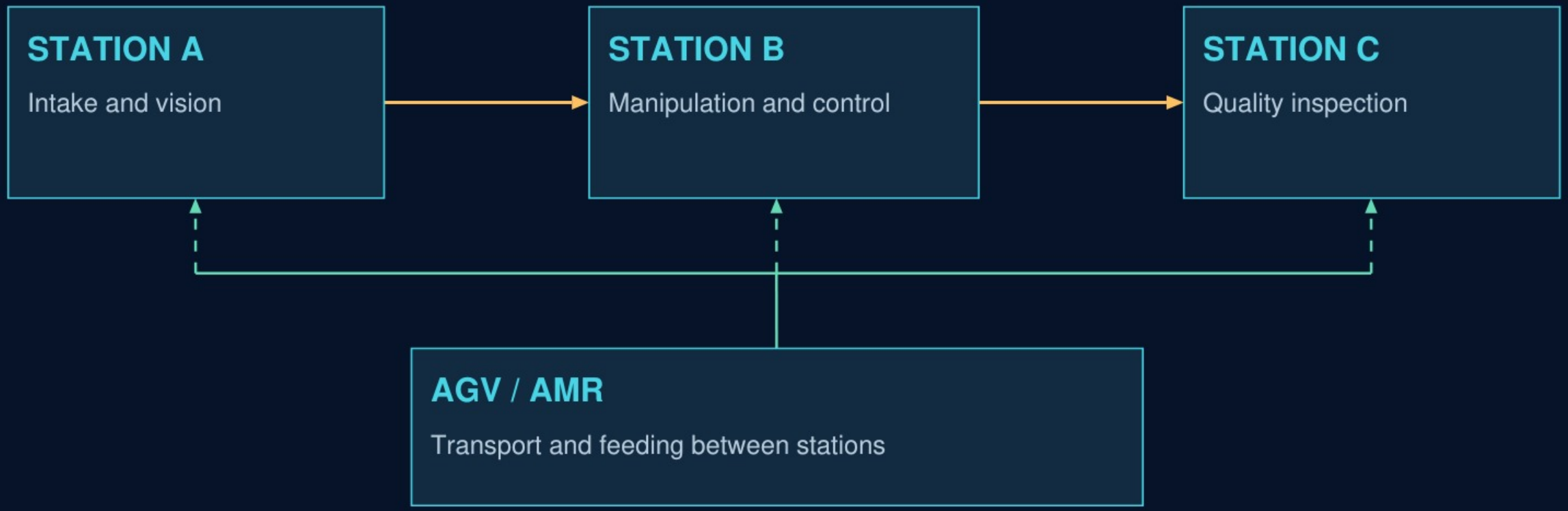
When does automation improve operations?

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Robots within an industrial cell

Physical and digital manufacturing integration



Logistics supports the process. Improving a cell need not raise total plant output.

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Fleet coordination

Scheduling orders, resources and routes



Data returns to assignment so the fleet can replan.

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Predictive maintenance

Availability is a critical economic variable

01

Sensors

Vibration, temperature and current.

02

Detection

Anomalies, rules and degradation.

03

Decision

Maintenance scheduling.

04

Outcome

Fewer unexpected failures and greater uptime.

An algorithm creates value when it prevents measurable operating losses.

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Human–robot collaboration

Automation and labour substitution are different questions

01

Substitution

Execute a specified task with risk controls.

02

Complementarity

Redesign jobs, improve ergonomics and support quality control.

03

Reorganisation

Changes in training, supervision and logistics.

Employment effects depend on tasks, institutions and worker reassignment.

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Process flow within a plant

Processes, buffers, safety and traceability



Buffers, traceability and safety require a more detailed model.

The laboratory summarises capacities and quality, without simulating queues or paths.

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BLOCK 06 / 07

Automation economics

Productivity, investment returns and learning

Which variables make robot adoption economically viable?

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Total cost of ownership (TCO)

Costs beyond the purchase price

01

CAPEX

Robot, cell, tools and integration.

02

OPEX

Energy, licences, maintenance and staff.

03

Risks

Downtime, scrap, reconfiguration and safety.

04

Value

Quality, productive hours and flexibility.

Cost per good part is more informative than the robot purchase price.

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OEE and effective capacity

Three components of operating effectiveness

AVAILABILITY

Operating time / planned time

×

PERFORMANCE

Ideal cycle × total count / operating time

×

QUALITY

Good count / total count

OEE = A × P × Q

Example: 0.87 × 0.90 × 0.92 = 0.72036 ≈ 72.0%

The laboratory operating index is a synthetic proxy, distinct from measured OEE.

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Investment returns depend on context

Incremental cost compared with operating savings

A

Measurable benefits

Less rework, improved safety, productivity, quality and responsiveness.

B

Hidden costs

MES integration, training, initial downtime, maintenance and safety.

Evaluation needs scenarios, a cost of capital and sensitivity analysis.

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Robot density and productivity

Correlation does not establish causation

01

Sector

Automotive, electronics and food production involve different tasks.

02

Size

Investment capacity and scale affect adoption.

03

Institutions

Skills, wages and regulation influence returns.

Compare tasks and multifactor productivity alongside robot installation counts.

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Scenario laboratory

Technical parameters and economic KPIs

Throughput

Good parts per shift.

Capacity use

Share of available robot capacity used.

Unit cost

Operating cost per good part.

Transparent assumptions make model outputs interpretable.

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BLOCK 07 / 07

Evaluation, safety and future

Demonstrations and reliable deployment

What evidence would establish a transformation in robotics?

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Physical safety and cybersecurity

Converging risks in robots with AI

01

Physical

Collisions, pinch points and stability.

02

Perception

Occlusion, false negatives and calibration.

03

Digital

Remote access, data and updates.

04

Human

Supervision, training and procedures.

AI policies require physical safeguards and validated safety functions.

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Evaluating embodied robots

Benchmarks with multiple dimensions

1	**Define the task**	Environment, tolerances and success condition.
2	**Measure**	Success, failures, time and assistance.
3	**Perturb**	New objects, failures and people nearby.
4	**Repeat**	Reproducibility and statistical confidence.
5	**Audit**	Logs, incidents and traceability.

An isolated demonstration is not a representative benchmark.

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Standards and governance

Machinery, collaborative robots and data protection

01

Engineering

Risk analysis and safety functions.

02

Operations

Access rules, lockout and maintenance.

03

Data

Minimisation, storage and lawful processing.

Applicable requirements depend on the robot, application and jurisdiction.

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What could change by 2030

Conditional scenarios rather than predictions

01

Software

Generalist policies and adaptation.

02

Hardware

Dexterous hands, torque and batteries.

03

Production

Cost, support and field robustness.

04

Work

Task redesign and productivity.

Distinguish technological expectations from economic evidence.

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Five critical questions

A rigorous agenda for evaluating Chinese robotics

1	**Scale**	How many units are actually operating?
2	**Reliability**	What happens after thousands of operating hours?
3	**Economics**	What does each successful task cost?
4	**Safety**	How does the system respond to anomalies?
5	**Transfer**	Does it generalise to a different factory?

The goal is safe, reliable and economically useful systems.

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SYNTHESIS

Robotics as a technical and organisational system

Five layers for evaluating physical AI in industry

1	**Mechanics**	Energy, actuators, sensors and reliability.
2	**Algorithms**	Perception, planning and closed-loop control.
3	**Integration**	Tasks, buffers, MES and industrial logistics.
4	**Economics**	OEE, cost per part and investment returns.
5	**Governance**	Safety, measurement and traceability.

Primary sources for rigorous research

Complete URLs are in the speaker notes and audiovisual guide

1	**IFR World Robotics 2026**	Installations, suppliers and statistical distinctions.
2	**Unitree / UBTECH / AGIBOT**	Manufacturer documentation and demonstrations.
3	**DeepMind: RT-2 / Gemini**	VLA architecture and embodied reasoning.
4	**NIST / ISO**	Robot performance evaluation and safety.
5	**Technical documentation**	Models, control, simulation and industrial integration.